Project 2: Spectral Analysis of Alpha Power

General Instructions:

1. Self submission.
2. Submission date: 15.06.2026
3. The submission should include a report containing relevant code snippets for the main analyses and calculations, together with the figures and explanations for each question, as well as the full Colab notebook/script. There is no need to include in the report code snippets for data loading, plotting, or other technical steps that are not part of the actual analysis or calculations.
4. All figures should be briefly explained and interpreted, even if not explicitly requested in the question. The explanations and interpretations are more important than the figures and code snippets themselves. Make sure the figures contain appropriate titles, axes names and legends when needed.
5. Submit your report and colab notebook in Lamda.

Background

Electroencephalography (EEG) is a non-invasive neurophysiological technique that measures electrical activity generated by large populations of neurons in the brain using electrodes placed on the scalp. The recorded EEG signal contains neural oscillations across different frequency bands, including delta, theta, alpha, beta, and gamma rhythms, each associated with different brain states and cognitive processes. One of the most prominent rhythms during resting-state EEG is alpha activity (8–12 Hz), which is typically strongest over occipital and parietal regions and becomes more pronounced when the eyes are closed compared to when the eyes are open.

To study the spectral properties of EEG signals, researchers commonly transform the signal from the time domain into the frequency domain using spectral analysis methods such as the Fast Fourier Transform (FFT) and Power Spectral Density (PSD) estimation. These methods allow identification of dominant oscillatory frequencies and measurement of their power. A commonly used measure in resting-state EEG is the Individual Alpha Frequency (IAF), which represents the frequency within the alpha range that shows the maximal power for a specific participant. Because the exact alpha peak varies between individuals, the IAF can provide a personalized characterization of neural oscillatory activity.

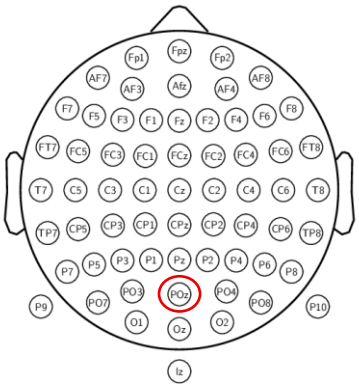
In this project, EEG recordings collected during Eyes Open (EO) and Eyes Closed (EC) resting-state conditions will be analyzed in order to compare different spectral analysis approaches and estimate alpha activity and IAF for multiple participants. Spectral methods including FFT, smoothed FFT, Welch PSD, spectrogram analysis, and signal reconstruction using IFFT will be used to examine how alpha oscillations appear in both the time and frequency domains.

Experiment Overview

EEG was recorded using the BioSemi system (containing 64 electrodes) from participants during two resting-state conditions:

- **Eyes Open (EO)** - participants were instructed to watch a fixation cross in the center of the screen for 1 minute.
- **Eyes Closed (EC) –** participants were instructed to close their eyes for 1 minute.

You will analyze recordings from electrode **POz** (red circle on the right image) from participants: **504, 519, 524.**



The recordings:

- Sampled at **fs = 1024 Hz**
- Contain **40 seconds** of data (the first and last 10 seconds were cut during pre-processing).

Part 1: Loading and Visualizing the EEG Signal

1. Load the EEG recordings for all participants and conditions.
2. Create a time vector spanning 0–40 seconds.
3. Plot the raw EEG signals for EO and EC overlayed for each participant. Choose an appropriate time window for visualizing the signal clearly.
4. Compare visually between EO and EC:
    - Which condition appears more rhythmic?
    - Do you notice visible alpha oscillations?

Part 2: Band-Pass Filtering (2–40 Hz)

1. **Apply** a 2–40 Hz band-pass filter to the EEG signal.
2. Use butterworth filter (function: scipy.signal.butter) and apply it to both EO and EC signals (function: scipy.signal.filtfilt). *This is a crucial step in processing EEG signals, but will be taught and applied in the next project.*
3. **Plot** a 2-second segment of the EEG signal for both EO and EC conditions. For each condition, overlay the raw signal and the band-pass filtered signal on the same plot.

Part 3: Spectral Analysis and IAF

This section is split to three: **Calculations**, **Figures**, **Discussion**.

You will compute 3 spectral estimates of the EEG signal and compare their properties.

Calculations

For each analysis, create a separate function. Make sure you use the filtered signals from Part 2, and apply all analyses to both EO and EC conditions.

1. **Power Spectrum using FFT**

Compute the power spectrum using the **Fast Fourier Transform (FFT)**.
Steps:

- Compute FFT of the filtered signal
- Comput the power spectrum.

2. **Smoothed FFT**

After computing the power spectrum using FFT apply a smoothing procedure across the frequency axis.
You may choose one smoothing method of your choice.
Describe in your report:

- Which smoothing method you used
- The smoothing parameter (for example window size or kernel width)

3. **PSD using Welch method**

Estimate the power spectrum using the **Welch method**.
Choose and report (and explain your choices):

- window type
- window length
- overlap

4. **Individual Alpha Frequency (IAF)**

Create a function that estimated the alpha peak frequency for each subject and each condition (EC and EO). Steps:

1. Define the alpha search range: 7.5-12.5 Hz.
2. Identify the frequency with maximum power within this range.
3. Record the IAF value for both EC and EO (report in the results).

Apply the function on each of the three spectral estimates calculated previously.

Figures

- For each participant, create a single figure containing three subplots:

    - FFT power spectrum
    - Smoothed FFT spectrum
    - Welch PSD
- In each subplot, overlay the EO and EC spectra.
- Mark the estimated IAF peak for each condition on the corresponding spectrum (for example using a scatter point or vertical line).
- Include the estimated IAF values in the figure, for example as part of the legend or subplot annotation.

Discussion

In your discussion, relate to:

- The differences between FFT, smoothed FFT, and Welch PSD spectra.
  Describe how the appearance of the spectra changes between methods, and explain why some methods may produce smoother or more interpretable spectral estimates than others.
- The visibility and sharpness of the alpha peak across methods.
  Discuss how smoothing and averaging influence the ability to identify dominant oscillatory activity and estimate the IAF.
- Differences between Eyes Open (EO) and Eyes Closed (EC) conditions.
  Describe how alpha activity changes between conditions and whether these changes are similar across participants.
- Individual differences between participants.
  Discuss possible reasons why spectral profiles and alpha activity may differ between individuals, and what this suggests about variability in brain activity.
- The estimated Individual Alpha Frequency (IAF).
  Compare the IAF values obtained from the different methods and discuss which method appears to provide the most stable or reliable estimate. Explain.

Part 4: Spectrogram

1. **Compute** a spectrogram of the EEG signal.
2. Explain:
    - What a spectrogram represents
    - How it differs from the power spectrum
3. Justify the spectrogram parameters you selected.
4. **Plot** the spectrogram per participant for EO and EC.
5. **Discuss** what the spectrogram reveals about alpha activity over time.

Part 5: Alpha Reconstruction Using IFFT

Create a function using the IAF estimate and the FFT signal produced in Part 3.

1. Choose the most reliable IAF estimate for each participant that you discussed.

2. Define an individual alpha band:
   IAF ±1 Hz
3. Keep only frequencies within this alpha range, and set all other frequencies to zero.
4. Reconstruct the signal using inverse FFT (IFFT).
5. For each participant, **plot** the reconstructed alpha signal overlayed with the original signal for both EC and EO. Choose an appropriate time window for visualizing the signal clearly.
6. **Compare** the reconstructed alpha signal to the original EEG signal.
   Discuss how isolating only the individual alpha band changes the appearance of the signal, what information is retained or removed by the reconstruction process, and whether differences between EO and EC conditions are visible in the reconstructed alpha activity.